ZNF292-Related Neurodevelopmental Disorder (Intellectual Developmental Disorder, Autosomal Dominant 64 / MRD64)

Comprehensive Disease Characteristics Report MONDO:0030934 · OMIM #619188 · Orphanet ORPHA:178469 · Category: Mendelian (autosomal dominant)

Summary

ZNF292-related neurodevelopmental disorder — formally **Intellectual Developmental Disorder, Autosomal Dominant 64 (MRD64; OMIM #619188)** — is a rare Mendelian condition caused by heterozygous, predominantly *de novo* **loss-of-function (truncating) variants in ZNF292**, a highly conserved C2H2 zinc-finger transcription factor on chromosome 6q14.3. The disorder was delineated as a discrete entity by Mirzaa and colleagues in 2020 from a cohort of 28 families, and the gene-disease relationship is now curated as **Definitive** by ClinGen/GenCC. The operative disease mechanism is **haploinsufficiency**: ZNF292 is among the most loss-of-function–intolerant genes in the human genome (gnomAD pLI = 1.0, LOEUF = 0.16), so a single inactivating allele is sufficient to cause disease.

Clinically, MRD64 is dominated by a neurodevelopmental triad — **intellectual disability/global developmental delay (~96%), speech and language delay (~93%), and autism spectrum disorder features (~55–70%)** — accompanied by a variable set of features including motor delay/hypotonia, ADHD, seizures, short stature, nonspecific facial dysmorphism, ocular abnormalities, feeding difficulties, and nonspecific brain MRI findings (e.g., ventriculomegaly). The condition has **low clinical recognizability** — there is no distinctive syndromic gestalt — so diagnosis is essentially molecular, achieved through trio exome/genome sequencing or multi-gene ID/ASD panels. Variants cluster in the large terminal exon 8, escape nonsense-mediated decay, and are predicted to produce a stable truncated protein.

Mechanistic work in human pluripotent stem-cell (hPSC)-derived cortical interneuron models (Kaushik et al., *Cell Reports* 2025) shows that ZNF292 exerts **stage-specific transcriptional control over GABAergic interneuron development**: it restrains differentiation genes in progenitors and promotes maturation genes in neurons. Its loss causes precocious progenitor differentiation, impaired interneuron maturation, altered ion-channel activity, elevated GABA responsiveness, and neuronal hyperactivity — pointing toward an **excitatory/inhibitory (E/I) imbalance** as the proximate cause of the neurodevelopmental phenotype. There is **no targeted or curative therapy**; management is entirely supportive and manifestation-based. Recurrence risk for siblings of a *de novo* proband is low (~1%, from possible parental germline mosaicism), while an affected individual has a 50% transmission risk.

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1. Disease Information

Overview. MRD64 is an autosomal dominant intellectual developmental disorder characterized by mildly to severely impaired intellectual development with prominent speech delay, frequently accompanied by autism spectrum disorder. It was formally defined by **Mirzaa et al., 2020** (*De novo and inherited variants in ZNF292 underlie a neurodevelopmental disorder with features of autism spectrum disorder*, *Genet Med* 22(3):538–546; PMID: 31723249; DOI 10.1038/s41436-019-0693-9).

Key identifiers.

Resource	Identifier
OMIM (phenotype)	**#619188** (Intellectual developmental disorder, autosomal dominant 64)
OMIM (gene)	*616213 (ZNF292)
Mondo	**MONDO:0030934**
Orphanet	ORPHA:178469; ORPHA:528084
Disease Ontology	DOID:0061037
HGNC	HGNC:18410
NCBI Gene	23036
Ensembl	ENSG00000188994
MANE Select transcript	NM_015021.3 (ENST00000369577.8)
Cytoband	6q14.3 (GRCh38 chr6:87,155,565–87,265,943)
ICD-11 / MeSH	No gene-specific code; coded under broad "disorders of intellectual development" / ASD categories; MeSH uses the gene term ZNF292

Synonyms / alternative names. MRD64; ZNF292-related neurodevelopmental disorder; ZNF292-related intellectual disability. Gene aliases: Zn-15, Zn-16, ZFP292, KIAA0530.

Information source. Derived largely from **aggregated disease-level resources and published patient cohorts** (OMIM, Orphanet, ClinVar/DECIPHER/DDD, and primary case series) rather than individual EHR records. The foundational cohort is the 28-family series of Mirzaa 2020, supplemented by later case reports and a 2026 series of 8 individuals (PMID: 42650143).

2. Etiology

Disease causal factors. The cause is **genetic and monogenic**: heterozygous loss-of-function variants in *ZNF292*. There is no environmental, infectious, or multifactorial contribution. In the defining cohort, **27 of 28 families carried *de novo* variants** and 1 was inherited from a mildly affected mother (Mirzaa 2020, PMID: 31723249).

Genetic risk factors. The causal variants themselves are the risk factor — predominantly frameshift and nonsense truncating variants. ZNF292 is **extraordinarily loss-of-function–intolerant** (gnomAD pLI = 1.0, LOEUF = 0.16; DECIPHER %HI = 26.56), so a single inactivating allele confers disease. No separate susceptibility loci or validated modifier genes have been established.

Environmental risk factors. None identified. Advanced parental age is a general (nonspecific) contributor to *de novo* mutation burden but has not been specifically linked to MRD64.

Protective factors. No genetic or environmental protective factors are known. The presence of 5 predicted-LoF ZNF292 variants in gnomAD population controls, and a family with an unaffected transmitting father of three affected siblings, suggests **rare incomplete penetrance**, implying that unknown genetic/stochastic modifiers can buffer the phenotype — but no specific protective allele has been identified.

Gene–environment interactions. No established GxE interactions. The disorder is a cell-autonomous developmental consequence of ZNF292 dosage reduction.

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3. Phenotypes

The core phenotype is a neurodevelopmental triad with variable syndromic add-ons. Feature frequencies below derive primarily from the Mirzaa 2020 cohort (n = 28 families) and later series.

Phenotype	HPO term	Frequency	Type
Intellectual disability / global developmental delay	HP:0001249 / HP:0001263	~96%	Cognitive
Delayed speech & language development	HP:0000750	~93%	Cognitive/ behavioral
Autistic behavior / ASD	HP:0000729	~55–70% (≈61%)	Behavioral
Motor delay / hypotonia	HP:0001252	≥30%	Clinical sign
ADHD	HP:0007018	Variable	Behavioral
Seizures / epilepsy	HP:0001250	Variable minority	Clinical sign
Brain MRI abnormality (e.g., ventriculomegaly)	HP:0002119	Variable	Imaging
Microcephaly	HP:0000252	Minority	Physical
Short stature	HP:0004322	Variable–high*	Physical
Feeding difficulties	HP:0011968	Variable	Clinical sign
Constipation	HP:0002019	Variable	Clinical sign
Strabismus / nystagmus	HP:0000486 / HP:0000639	Minority	Clinical sign
Facial dysmorphism (thin upper lip, smooth philtrum, upturned nasal tip, synophrys, epicanthus, prominent forehead)	—	Nonspecific	Physical

*A 2026 series of 8 individuals (PMID: 42650143) found **short stature in all 8**, suggesting growth failure may be more consistent than previously appreciated. Importantly, IGF-1 is typically normal, distinguishing this from growth hormone deficiency.

Phenotype characteristics. - **Age of onset:** congenital/neurodevelopmental; delays recognized in infancy to early childhood. - **Severity:** variable (mild to severe). - **Progression:** stable/static — a non-progressive (static) encephalopathy; features are lifelong but do not degenerate. - **Expressivity:** highly variable even for identical recurrent variants (see Section 9).

Quality-of-life impact. Lifelong cognitive, communicative, and behavioral impairment produces substantial effects on daily functioning, educational attainment, and independence. No disease-specific QoL instrument (EQ-5D/SF-36/PROMIS) data exist; impact is inferred from the ID/ASD phenotype.

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4. Genetic / Molecular Information

Causal gene. *ZNF292* (OMIM *616213; HGNC:18410; NCBI Gene 23036; Ensembl ENSG00000188994), 6q14.3. Encodes a 2,192-amino-acid, ~156 kDa Krüppel C2H2-type zinc-finger transcription factor with 15–16 C2H2 zinc fingers, N-terminal tandem TFIIIA-type fingers, long C-terminal linkers enabling cooperative DNA binding, 2–3 nuclear localization signals, and coiled-coil domains. UniProt O60281** (ZN292_HUMAN); AlphaFold model AF-O60281-F7; InterPro IPR013087 / Pfam PF00096 (zf-C2H2). Paralog: RLF.

Pathogenic variants. - **Classification:** Pathogenic / Likely Pathogenic truncating variants per ACMG/AMP. ClinVar holds roughly **44 pathogenic + 54 likely pathogenic** alleles; DECIPHER/DDD lists >13 P/LP entries. - **Variant type/class:** Predominantly **frameshift and nonsense** (truncating). In Mirzaa 2020: 24 distinct heterozygous nonsense/frameshift variants across 28 patients — **19 frameshift + 7 nonsense (26 of 28)** residing in **terminal exon 8**, plus 1 *de novo* frameshift in exon 4 and 1 *de novo* nonsense in exon 2. - **NMD escape:** Exon 8 encodes all 16 C2H2 zinc-finger domains (~85% of the coding sequence). Because variants fall in the last exon, they **largely escape nonsense-mediated decay** and produce a stable truncated protein (RT-PCR confirmed stable transcript from both alleles). - **Recurrent variants:** c.3066_3069delAAGA (p.Glu1022AspfsTer3; NM_015021.2; ClinVar Variation ID 431106, Pathogenic); and c.6160_6161del (p.Glu2054Lysfs*14), now the most frequently observed variant (7 individuals across unrelated families, PMID: 42650143). Newer variants include c.4189C>T (p.Arg1397Ter), c.6343C>T (p.Arg2115Ter), c.1533del (p.Ile511Metfs*11), c.3094dup (p.Ser1032Phefs*18), c.5977_5978del (p.Gln1993fs). - Allele frequency: All but one variant absent from gnomAD v2.1; variants are private/ultra-rare. - Origin: Germline, overwhelmingly* de novo*. (Somatic ZNF292 mutations occur in gastric/colorectal cancer and CLL — a separate, non-germline tumor-suppressor context, not MRD64.) - Functional consequence: Loss of function via haploinsufficiency.**

Modifier genes. None validated. Incomplete penetrance in rare families implies unidentified modifiers.

Epigenetic information. No validated ZNF292-specific DNA-methylation episignature exists in the published record (not part of the EpiSign classifier).

Chromosomal abnormalities. Whole-gene **6q14.3 deletions encompassing ZNF292** have been reported in ASD cohorts; detectable by chromosomal microarray.

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5. Environmental Information

- **Environmental factors:** None implicated. MRD64 is a purely genetic, cell-autonomous developmental disorder.
- **Lifestyle factors:** Not applicable.
- **Infectious agents:** Not applicable.

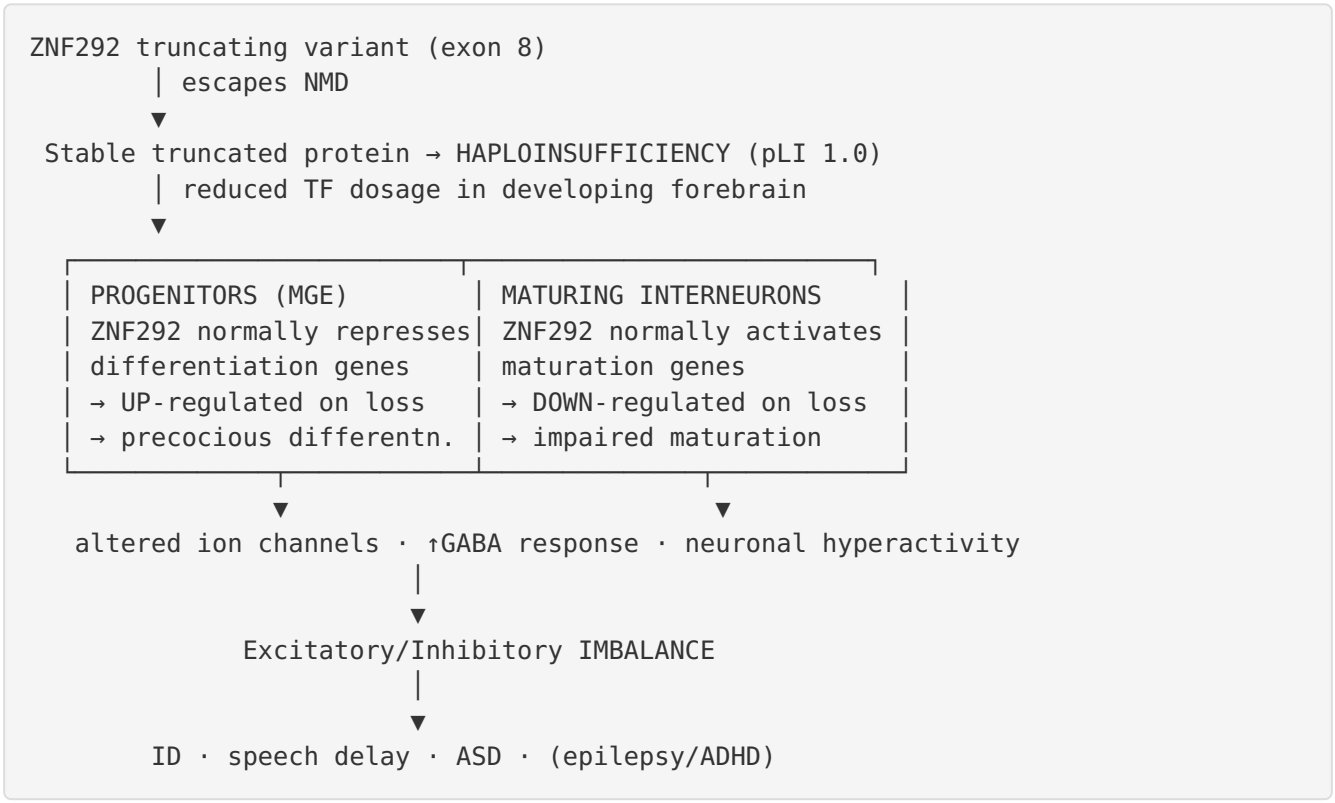
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6. Mechanism / Pathophysiology

Ordered causal chain

1. A **heterozygous truncating variant** in *ZNF292* (frameshift/nonsense, usually in terminal exon 8) arises *de novo* in the germline → **leads to** a transcript that largely **escapes nonsense-mediated decay**.
2. NMD escape **results in** a **stable truncated ZNF292 protein** lacking normal function; combined with the normal allele this **produces haploinsufficiency** (ZNF292 is pLI = 1.0 / LOEUF = 0.16, i.e., dosage-sensitive). *[Loss of function is the demonstrated mechanism; whether the truncated protein also exerts a dominant-negative effect is not established — inferred open question.]*
3. Reduced functional ZNF292 dosage in the developing forebrain **disrupts stage-specific transcriptional programs** that ZNF292 normally governs (ZNF292 is a nuclear DNA-binding TF, GO:0005634 nucleus; biological process GO:0006355 regulation of transcription).
4. **Branch A — progenitors (medial ganglionic eminence / ventral telencephalon, UBERON:0002264):** ZNF292 normally **restrains** differentiation, neuronal morphogenesis/axon-guidance, and synapse/ion-channel genes (direct targets including NKX2.1, ASCL1, SATB1, WNT5B). Its loss **up-regulates** these targets → **precocious differentiation** of GABAergic interneuron progenitors.

5. **Branch B — maturing neurons (cortical interneurons, CL:0000617 GABAergic interneuron):** ZNF292 normally **promotes** interneuron maturation genes (including other NDD genes). Loss reduces ZNF292 genome-wide binding and **down-regulates** these targets → **compromised interneuron maturation/function**.
6. These converge to **altered ion-channel activity, elevated GABA responsiveness, and neuronal hyperactivity → excitatory/inhibitory (E/I) imbalance** in cortical circuits.
7. Disrupted GABAergic interneuron development and E/I imbalance **result in** the clinical phenotype: intellectual disability, speech/language delay, autism spectrum features, and variable epilepsy/ADHD. *[The link from stem-cell phenotypes to human clinical features is inferred, not directly demonstrated in patients.]*



Detail by category

- **Molecular pathways / cellular processes:** Transcriptional regulation of neural differentiation and synaptogenesis. ZNF292 binding sites are enriched for **bHLH, homeodomain, and Ets** transcription-factor motifs, indicating cooperative transcriptional networks. Processes: neuron differentiation (GO:0030182), interneuron migration/ maturation, synapse formation, ion-channel organization.

- **Protein dysfunction:** Loss of function of a C2H2 zinc-finger DNA-binding protein; truncation removes/disrupts the zinc-finger DNA-binding array encoded by exon 8.
- **Cell types involved:** GABAergic cortical interneurons (CL:0000617), interneurons (CL:0000099), and neuronal stem/progenitor cells (CL:0000047), originating in the medial ganglionic eminence.
- **Biochemical/electrophysiological abnormality:** Altered ion-channel activities, elevated GABA responsiveness, and neuronal hyperactivity in ZNF292-deficient interneurons.
- **Molecular profiling evidence:** Genome-wide ZNF292 occupancy plus transcriptomics in CRISPRi-knockdown hPSC-derived MGE-like NPCs (D15) and cortical interneurons (D30) (Kaushik et al., *Cell Reports* 2025 May 27;44(5):115597; PMID: 40257863). Human ZNF292 expression peaks ~12 weeks post-conception and continues lifelong.
- **Evolutionary/functional context:** ZNF292 is the human ortholog of rat Zn-15 / mouse Zn-16, originally characterized as a **growth-hormone-gene transcriptional activator cooperating with POU1F1/Pit-1 (GHF-1)** — a plausible (inferred) link to the short-stature phenotype via the GH axis.

Upstream vs downstream: The mutation and haploinsufficiency are the most upstream events; transcriptional dysregulation of interneuron programs is intermediate; altered channel physiology, GABAergic hyperexcitability, and E/I imbalance are downstream; clinical neurodevelopmental features are the terminal manifestation.

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7. Anatomical Structures Affected

- **Primary organ/system:** Brain / central nervous system — UBERON:0000955 (brain), UBERON:0000956 (cerebral cortex); developmental origin UBERON:0002264 (ventral telencephalon / medial ganglionic eminence). Body system: **nervous system**.
- **Tissue/cell level:** Nervous tissue; GABAergic interneurons (CL:0000617), interneurons (CL:0000099), neuronal progenitors (CL:0000047).
- **Subcellular level:** Nucleus (GO:0005634); ZNF292 localizes to nucleus/nucleoplasm/nuclear membrane and acts as a DNA-binding transcription factor.
- **Secondary / variable involvement:** Endocrine (pituitary/GH axis → short stature; UBERON:0000007 pituitary gland), musculoskeletal (short stature, hypotonia), visual system (strabismus/nystagmus; UBERON:0000970 eye), digestive system (feeding difficulties, constipation; UBERON:0001555 digestive tract), occasional cardiac involvement.

- **Lateralization:** Bilateral / systemic — not lateralized.

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8. Temporal Development

- **Onset:** Congenital/neurodevelopmental; delays recognized in infancy to early childhood. Onset pattern is **chronic/insidious** (developmental), not acute.
- **Progression: Static (non-progressive)** encephalopathy — features are stable and lifelong rather than degenerative.
- **Disease course:** Chronic, lifelong.
- **Remission:** No spontaneous remission; developmental gains occur with intervention but underlying impairment persists.
- **Critical periods:** The vulnerable window is prenatal/early-postnatal corticogenesis (ZNF292 expression peaks ~12 weeks post-conception), when interneuron specification and maturation occur — the putative window for any future intervention.

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9. Inheritance and Population

- **Inheritance pattern: Autosomal dominant** (OMIM #619188), predominantly *de novo* (27/28 families in Mirzaa 2020). Rare transmitted cases exist (1 family, variant from a mildly affected mother).
- **Penetrance:** High but **not complete** — 5 predicted-LoF ZNF292 variants in gnomAD controls and one unaffected transmitting father of three affected siblings indicate rare non-penetrance.
- **Expressivity: Highly variable** — the recurrent c.6160_6161del (p.Glu2054Lysfs*14) produces presentations from severe global DD to mild neurodevelopmental impairment (PMID: 42650143).
- **Anticipation / germline mosaicism:** No anticipation (not a repeat-expansion disorder). Parental germline mosaicism is the basis for the low (~1%) sibling recurrence risk in *de novo* cases.
- **Founder effects / consanguinity / carrier frequency:** None — dominant, mostly *de novo*; carrier screening not applicable.

- **Epidemiology:** No dedicated Orphanet prevalence or life-expectancy estimate; recently described and rare.
- **Demographics:** No strong sex bias established; no ethnic or geographic clustering reported. Age distribution: diagnosed in childhood.

Feature frequencies (n = 28): DD/ID 96%; speech delay 93%; ASD/autistic features ~61%; ADHD, tone abnormalities, structural brain abnormalities, dysmorphism, ocular features, and growth failure each in ≥30%.

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10. Diagnostics

- **Primary approach:** Molecular — diagnosis is **sequencing-driven** given low clinical recognizability. Trio **exome sequencing (ES)**, **genome sequencing (WGS)**, or multi-gene **ID/ASD NGS panels**, with Sanger confirmation of *de novo* status in proband and parents (Mirzaa 2020, PMID: 31723249; 2026 series used WES in 8 unrelated individuals finding 7 *de novo* variants, PMID: 42650143). ZNF292 is a strong candidate in exome-negative idiopathic DD/ID/ASD.
- **Chromosomal microarray (CMA):** Detects 6q14.3 deletions encompassing ZNF292.
- **Biomarkers / labs:** No specific biomarker or metabolic test. IGF-1 is typically **normal** (distinguishes from GH deficiency despite short stature).
- **Imaging:** Brain MRI findings are nonspecific (ventriculomegaly, variable).
- **Electrophysiology:** No disease-specific EEG protocol; standard epilepsy workup applies when seizures occur.
- **Omics diagnostics / episignature:** No validated DNA-methylation episignature.
- **Clinical criteria:** No formal diagnostic criteria; molecular confirmation defines the diagnosis.
- **Differential diagnosis:** Low syndromic recognizability — distinguish from other monogenic ID/ASD causes, childhood apraxia of speech/oral-motor apraxia/velopharyngeal insufficiency, Noonan syndrome (ZNF292 lacks cardiac defects/typical facies), GH deficiency (normal IGF-1 in ZNF292), and Turner syndrome (via karyotype/CMA).
- **Screening:** Not part of newborn or carrier screening panels; cascade testing recommended given rare inherited cases and incomplete penetrance.

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11. Outcome / Prognosis

- **Survival / mortality:** No evidence of reduced life expectancy; not a life-limiting condition per se (no disease-specific mortality data).
- **Morbidity / function:** Lifelong cognitive, communicative, and behavioral disability is the main burden; degree correlates with ID severity.
- **Disease course:** Static (non-progressive); developmental progress achievable with intervention.
- **Complications:** Seizures, ADHD, feeding/growth issues, and ophthalmologic problems when present.
- **Prognostic factors:** Severity of baseline ID, presence of epilepsy, and access to early intervention; no validated molecular prognostic biomarker (genotype–phenotype correlation is weak, given variable expressivity of recurrent variants).
- **Quality of life:** No disease-specific QoL measures published.

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12. Treatment

There is no disease-specific, targeted, or curative therapy. Management is **supportive and multidisciplinary**, directed at individual manifestations:

- **Developmental / rehabilitative:** Early intervention, special education, **speech-language therapy** (NCIT: Speech Therapy), occupational therapy, physical therapy.
- **Behavioral:** ASD behavioral therapy; ADHD management.
- **Pharmacotherapy (symptomatic):** Standard antiseizure medications for epilepsy; ADHD pharmacotherapy as indicated. No pharmacogenomic considerations specific to ZNF292.
- **Monitoring:** Growth/feeding surveillance; ophthalmologic monitoring (strabismus/nystagmus).
- **Advanced / experimental therapeutics:** None approved or in trials specific to ZNF292 (no gene, cell, or RNA-based therapy).
- **Personalized medicine:** Not yet applicable beyond molecular diagnosis.

NCIT suggestions where applicable: Speech Therapy, Occupational Therapy, Physical Therapy, Behavioral Therapy, Supportive Care.

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13. Prevention

- **Primary prevention:** None possible (de novo genetic etiology; no environmental cause).
- **Secondary/tertiary prevention:** Early developmental intervention and surveillance for complications (seizures, growth, vision) mitigate functional impact.
- **Genetic counseling:** For a proband with a *de novo* variant, sibling recurrence risk is low (~1%, from possible parental germline mosaicism); an affected individual has a **50% transmission risk** (AD). Variable expressivity and rare incomplete penetrance should be discussed.
- **Prenatal / preimplantation testing:** PGT-M and prenatal diagnosis are possible once a familial variant is identified.
- **Screening programs / immunization / public health:** Not applicable; not part of newborn or carrier screening.

14. Other Species / Natural Disease

- **Orthologs (DIOPT):** Mouse *Zfp292* (MGI:1353423; NCBI Gene 30046; chr4; aliases Zn-15/Zn-16/Zfp-15/Krox-10; score 16/16); rat *Zfp292* (RGD 11811932; 14/14); zebrafish duplicates *znf292a* (ZFIN ZDB-GENE-071004-18) and *znf292b* (ZFIN ZDB-GENE-030131-9559; best fish ortholog 13/15). **No Drosophila ortholog** — the gene is vertebrate-restricted (1 human, 1 mouse, 1 rat, 2 zebrafish).
- **Natural disease in other species:** No naturally occurring ZNF292-related disease has been catalogued (no OMIA entry noted).
- **Comparative biology:** The rat Zn-15 / mouse Zn-16 ortholog was characterized as a growth-hormone-gene transcriptional activator cooperating with Pit-1/POU1F1 — evolutionarily conserved transcriptional function.
- **Transmission / zoonosis:** Not applicable.

15. Model Organisms

- **Principal disease model: Human pluripotent stem-cell (hPSC)-derived cortical interneuron systems** using CRISPRi (dCas9-KRAB) knockdown to model haploinsufficiency (Kaushik et al., *Cell Reports* 2025; PMID: 40257863). These differentiate to MGE-like ventral telencephalic NPCs (D15) and cortical interneurons (D30) and recapitulate haploinsufficiency-driven differentiation/maturation defects and GABAergic hyperexcitability.
- **Mouse resources:** IMPC/MGI lists ~75 alleles (gene-trap, targeted, endonuclease-mediated) across 151 strains, but **no published ZNF292-disease-specific mouse study recapitulating MRD64** was identified.
- **Model limitations:** Stem-cell models capture cell-autonomous developmental and electrophysiological phenotypes but not whole-organism behavior, cognition, or systemic (e.g., growth) features; no validated behavioral animal model exists.
- **Resources:** MGI, IMPC, ZFIN, RGD, Cellosaurus/ATCC (for hPSC lines).

Mechanistic Model / Interpretation

The data converge on a coherent model in which **ZNF292 is a dosage-sensitive master transcriptional regulator of cortical GABAergic interneuron development**. Its defining genetic feature — truncating variants in terminal exon 8 that escape NMD — is unusual in that it produces a *stable* truncated protein, yet the functional and population-genetic evidence (pLI = 1.0, LOEUF = 0.16) indicate the operative mechanism is **loss of function through haploinsufficiency** rather than a classic degraded null. Whether the truncated protein additionally exerts a dominant-negative effect remains an open, testable question.

Functionally, ZNF292 acts in a **stage-dependent, bidirectional manner**: a transcriptional brake on differentiation genes in progenitors, and an activator of maturation genes in neurons. Removing one allele's worth of this regulator simultaneously accelerates progenitor exit (precocious differentiation) and impairs the maturation program, yielding interneurons with altered ion-channel profiles, exaggerated GABA responses, and network hyperactivity. The downstream clinical correlate — **E/I imbalance** — is a recurring theme across monogenic ID/ASD/epilepsy disorders and plausibly links the molecular lesion to the ID–speech-delay–autism triad and the variable epilepsy/ADHD.

The **weak genotype–phenotype correlation** (identical recurrent variants producing severe-to-mild phenotypes) and **rare incomplete penetrance** argue that genetic background and stochastic developmental factors substantially shape outcome — a key consideration for counseling and for any future precision approach.

Evidence Base

PMID	Study	Type	Contribution
31723249	Mirzaa et al., *Genet Med* 2020;22(3):538–546	Human clinical cohort (28 families)	Defines MRD64; establishes *de novo* truncating LoF variants clustered in exon 8, NMD escape, AD inheritance, core phenotype frequencies
40257863	Kaushik et al., *Cell Reports* 2025;44(5):115597	hPSC / CRISPRi model + genomics	Stage-specific transcriptional control of cortical interneurons; precocious differentiation, impaired maturation, GABAergic hyperexcitability, direct targets and motif enrichment
42650143	2026 series (8 individuals)	Human clinical series	Expands phenotype (short stature in all 8), recurrent c.6160_6161del with variable expressivity, WES diagnostic yield

Supporting resource-level evidence: OMIM #619188 / *616213; gnomAD constraint metrics (pLI 1.0, LOEUF 0.16); ClinGen/GenCC Definitive classification; ClinVar (~44 P + ~54 LP alleles); UniProt O60281; DIOPT ortholog scores; MGI/IMPC allele catalogs.

Limitations and Knowledge Gaps

1. **No animal model of the disease** recapitulating MRD64 — the mechanism rests heavily on a single hPSC/CRISPRi study; in vivo validation of E/I imbalance and behavioral correlates is lacking.
2. **LoF vs. dominant-negative** — stable truncated protein raises the possibility of a dominant-negative or neomorphic contribution that has not been experimentally excluded.

3. **Weak genotype–phenotype correlation and incomplete penetrance** — modifiers are unidentified; prognosis prediction is limited.
4. **Epidemiology is unquantified** — no prevalence, incidence, sex ratio, or natural-history/survival data; cohorts are small.
5. **Short stature mechanism** — the GH-axis link (via the ortholog's role with POU1F1/Pit-1) is suggestive but unproven in patients; IGF-1 is normal.
6. **No biomarker or episignature** — diagnosis depends entirely on sequencing; no functional confirmatory assay.
7. **No therapeutics pipeline** — no targeted, gene, or RNA-based therapy exists or is in trials.

Proposed Follow-up Experiments / Actions

1. **Generate a conditional *Zfp292* mouse (and/or zebrafish) model** to test interneuron developmental defects, E/I balance, seizure susceptibility, cognition/social behavior, and growth in vivo.
2. **Distinguish LoF from dominant-negative mechanism** by comparing heterozygous knockout vs. truncation knock-in in isogenic hPSC lines and in vivo models.
3. **Establish a patient registry / natural-history study** to quantify prevalence, feature frequencies, genotype–phenotype correlations, and long-term outcomes; standardize QoL/functional measures.
4. **Characterize the GH-axis/short-stature link** (endocrine profiling across a larger cohort; test ZNF292 regulation of GH/POU1F1 targets in relevant cell types).
5. **Search for modifiers of penetrance/expressivity** via genome-wide analysis in discordant carriers (e.g., the unaffected transmitting father and mildly affected relatives).
6. **Evaluate E/I-rebalancing pharmacology** (e.g., agents modulating GABAergic hyperexcitability) in the hPSC interneuron model as a therapeutic-hypothesis screen.
7. **Assess for a DNA-methylation episignature** across a larger patient set to determine whether a confirmatory epigenetic classifier is feasible.

Report compiled from 11 confirmed findings across 5 investigation iterations. Evidence sources span human clinical cohorts (Mirzaa 2020; 2026 series), an in vitro hPSC/CRISPRi functional-genomics study (Kaushik 2025), and curated resource databases (OMIM, gnomAD, ClinGen/GenCC, ClinVar, UniProt, MGI/IMPC, DIOPT).
